

LEAF: Language-EEG Aligned Foundation Model for Brain-Computer Interfaces

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Abstract—Recent advances in electroencephalography (EEG) foundation models, which capture transferable EEG representations, have greatly accelerated the development of brain-computer interfaces (BCI). However, existing approaches still struggle to incorporate language instructions as prior constraints for EEG representation learning, limiting their ability to leverage the semantic knowledge inherent in language to unify different labels and tasks. To address this challenge, we present LEAF, a foundation model for EEG–Language Alignment with Semantic Task Instruction and Querying. LEAF integrates task-aware semantic guidance to produce structured and linguistically aligned EEG embeddings, thereby enhancing decoding robustness and transferability. In the EEG pretraining stage, we introduce a joint Spectral–Temporal Reconstruction (STR) framework that captures the coupled spectral rhythms and temporal dynamics of EEG signals. STR applies randomized spectral perturbation to enhance frequency robustness and uses two complementary temporal objectives to learn both contextual and sequential structure. In the EEG–Language alignment stage, we propose the Instruction-conditioned Q-Former (IQF). This query-based cross-attention transformer injects instruction embeddings into EEG tokens and achieves semantic alignment with textual label embeddings through learnable queries. We evaluate LEAF on 16 downstream datasets spanning motor imagery, emotion recognition, steady-state visual evoked potentials, covert speech, and healthcare tasks. LEAF achieves state-of-the-art performance on 12 of the 16 datasets and obtains the best average results across all five task categories. Importantly, our analyses reveal for the first time that explicit task instructions serve as semantic priors guiding EEG embeddings into coherent and linguistically grounded spaces. The code and pre-trained weights will be released.

Index Terms—Electroencephalography (EEG), brain-computer interface (BCI), foundation models, language alignment, semantic task instruction, motor imagery, and emotion recognition.

I. INTRODUCTION

Electroencephalography (EEG) provides noninvasive brain dynamics measurement with millisecond-level temporal resolution, making it particularly suitable for applications such as motor imagery (MI) decoding, emotion recognition, and

steady-state visual evoked potential (SSVEP) classification. In addition to its high temporal precision, EEG offers the advantages of portability, relatively low cost, and suitability for long-term monitoring. However, EEG suffers from low signal-to-noise ratio, nonstationarity, and large variability across subjects, datasets, and tasks, which has historically limited its generalizability [1]. These shortcomings motivate the development of EEG foundation models (EEG-FMs), which aim to leverage large-scale pretraining to learn transferable representations that can overcome variability and improve downstream task performance. Typically, EEGPT [2] applies transformer-based pretraining to capture temporal dependencies. LaBraM [3] leverages masked autoencoding on large EEG corpora to learn generalizable embeddings. CBraMod [4] focuses on cross-brain modeling to facilitate cross-subject transfer. However, most existing EEG-FMs are trained with discrete task labels (e.g., 0/1 rather than happy/angry), which limits their ability to exploit the richer semantic information associated with tasks and labels. The lack of explicit EEG–language coupling may therefore hinder generalization across heterogeneous downstream settings. More recently, Large Language Models (LLMs) have been introduced to further enhance EEG-FMs due to their tremendous success in natural language processing [5] and multimodal understanding [6]. As a pioneering work, NeuroLM [7] aligns EEG and language embeddings by training a text-aligned neural tokenizer. Specifically, EEG signals are discretized into tokens and adversarially forced into the same embedding space as text. These EEG tokens are then added to the LLM vocabulary and jointly modeled with text through multi-channel autoregression and instruction tuning. While promising, current EEG-FMs and EEG–language FMs still face two major limitations: First, **missing spectro-temporal interaction**: EEG signals exhibit strong coupling between spectral rhythms and multi-timescale temporal dynamics. These dynamics include causal transitions such as event-related potentials and contextual fluctuations reflecting cognitive or emotional states [8], [9]. However, existing methods do not jointly model spectral structure together with both contextual and causal temporal dependencies, making it difficult to capture coherent patterns that are essential for reliable EEG representation learning. Second, **insufficient semantic-level alignment**: current EEG–language FMs such as NeuroLM align EEG signals with language through a coarse-grained distribution matching objective. This approach does not incorporate fine-grained semantic constraints and therefore limits the ability of language for guiding EEG representations.

To address these two problems, we propose **LEAF**, a foun-

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dation model for EEG–Language Alignment with Semantic Task Instruction and Querying. This approach first introduces a **joint spectral–temporal reconstruction framework** that unifies frequency modeling with both bidirectional and causal temporal learning. By combining global spectral perturbation with complementary temporal masking strategies, the model learns frequency-aware and contextually rich EEG representations, laying a stronger foundation for downstream tasks. To further bridge EEG signals with semantic information, we propose an **Instruction-conditioned Q-Former (IQF)** that aligns EEG representations with natural language at the semantic level. Specifically, EEG embeddings are modulated by task-level instructions (e.g., “This is an MI task”, “Decode emotion from EEG”) and label semantics (e.g., “Left”, “Happy”), thereby guiding representation learning toward task-relevant dimensions. The modulated EEG features are then refined through cross-attention with learnable queries, enabling instruction-driven alignment between EEG and language representations.

Our main contributions are as follows:

- We introduce **LEAF**, a novel EEG–Language Foundation Model designed for EEG decoding across diverse BCI applications. LEAF unifies spectral–temporal modeling with instruction guidance to realize semantic-level EEG–language alignment, thereby enhancing transferability and interpretability across heterogeneous downstream tasks.
- We design two key components to realize this framework: a joint **Spectral–Temporal Reconstruction (STR)** module that jointly captures frequency along with both causal and contextual temporal dynamics, and an **Instruction-conditioned Q-Former (IQF)** that integrates task instructions and label semantics into EEG features through query-based cross-modal alignment.
- We conduct a comprehensive evaluation on 16 downstream EEG datasets spanning motor imagery, emotion, SSVEP, covert speech, and healthcare tasks. LEAF achieves state-of-the-art (SOTA) average performances across all 5 tasks and demonstrates strong generalization across datasets.
- For the first time, we demonstrate that explicit instructions act as semantic priors that restructure EEG feature spaces for better separability, and that stronger text encoders supply richer semantics, leading to faster convergence, higher accuracy, and improved generalization.

II. RELATED WORK

a) Self-supervised Pretraining.: Self-supervised pretraining has emerged as a powerful paradigm in representation learning, reducing the reliance on large amounts of annotated data while leveraging abundant unlabeled signals. Self-supervised methods design pretext tasks that encourage models to learn meaningful feature representations from the inherent structure of data. Early successes in natural language processing, such as BERT [10] and GPT series [11], demonstrated that masked language modeling and next-word prediction can yield representations transferable to diverse downstream tasks. Similarly, in computer vision, contrastive learning approaches like SimCLR [12], MoCo [13] and masked image modeling like

MAE [14] showed that pretraining on large-scale unlabeled images leads to robust and generalizable visual features.

b) EEG Foundation Models.: The concept of foundation models has recently expanded into the EEG domain, aiming to build large-scale pre-trained backbones that generalize across datasets, tasks, and clinical conditions. Several pioneering efforts have been proposed. BIOT [15] explored scalable transformer-based architectures for biomedical signals, positioning EEG as a central modality. EEGPT [2], inspired by advances in language modeling, leveraged transformer pretraining strategies such as masked prediction and contrastive learning to enhance generalization across heterogeneous EEG datasets. LaBraM [3] introduced a large-brain-model framework, emphasizing cross-dataset pretraining to capture universal EEG representations. CBraMod [4] extended this idea by focusing on cross-brain modularity, enabling adaptation across diverse cognitive and motor tasks. Beyond EEG-specific approaches, NeuroLM [7] proposed a broader neural language model for neuroscience data, while PhysioOmni [16] further expanded the scope to multi-physiological modalities, integrating EEG with signals such as ECG and EMG to learn cross-modal representations. Collectively, these efforts highlight the emerging trajectory of EEG-FMs: moving from task-specific networks toward unified, pre-trained architectures capable of powering downstream applications with minimal fine-tuning, and paving the way for general-purpose brain decoding systems.

III. METHOD

In this section, we introduce the design of **LEAF**, our proposed EEG–Language foundation model. LEAF is trained in two stages: an EEG pretraining stage, where a joint spectral–temporal objective encourages frequency-aware and temporally predictive EEG representations, and a multi-task instruction tuning stage, where EEG embeddings are conditioned on task instructions and aligned with textual targets to improve decoding performance across diverse tasks. An overview of the LEAF architecture is shown in Fig. 1.

A. Pretraining with Joint Spectral–Temporal Reconstruction

EEG signals combine spectral rhythms with temporal dynamics that exhibit both causal transitions and broader contextual fluctuations. To learn representations that capture these spectral–temporal relationships during pretraining, we propose a joint spectral-temporal reconstruction module (STR). STR is a two-stream pretraining architecture built upon a shared spectral backbone. The backbone introduces randomized frequency suppression to enforce spectral invariance, while two complementary temporal branches operate in parallel to capture contextual and sequential dependencies. Let $X \in \mathbb{R}^{C \times T}$ denote an EEG trial with C channels and T time points. To manage long recordings and improve training stability, X is segmented into a sequence of non-overlapping windows of fixed length t , yielding segments $x_i \in \mathbb{R}^{C \times t}$ for $i = 1, \dots, \lfloor T/t \rfloor$. Each segment captures synchronized activity across all channels within the temporal window. All EEG inputs are spatially interpolated to a standardized 65-channel

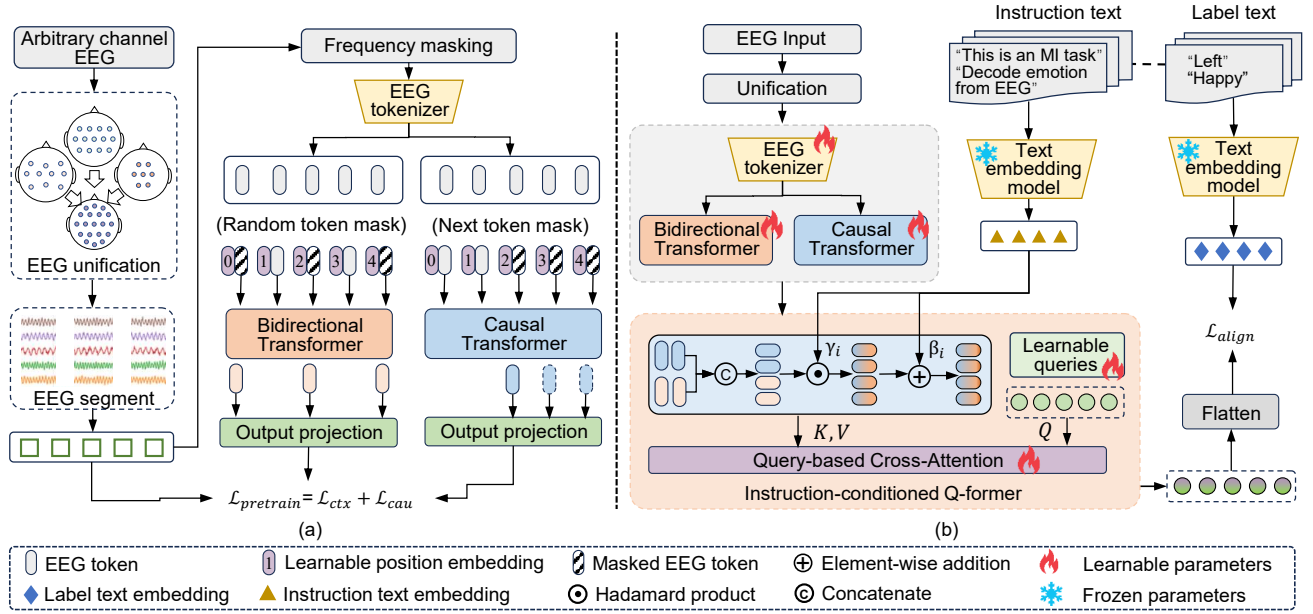


Fig. 1: The architecture design of LEAF. **(a)** joint Spectral-Temporal Reconstruction module (STR) for self-supervised EEG pretraining, combining frequency masking, global context modeling, and temporal sequence learning. **(b)** During multi-task instruction tuning, an Instruction-conditioned Q-Former (IQF) aligns EEG signals with language by injecting instruction embeddings and leveraging query-based cross-attention.

10–10 montage before tokenization; the electrode layout is provided in Appendix A.

a) Spectral masking backbone: The spectral masking backbone encourages the encoder to learn frequency-aware features across paradigms. We first suppress a randomly chosen frequency band in each segment before tokenization. For x_i , compute its spectrum $X_{f,i} = \text{FFT}(x_i)$ and randomly select a band $[f_{\min}, f_{\max}]$ of width f_{band} to remove, producing a masked spectrum $\tilde{X}_{f,i} = \mathcal{M}_{[f_{\min}, f_{\max}]}(X_{f,i})$. Then we conduct an inverse transform to get the perturbed signal via $\tilde{x}_i = \text{iFFT}(\tilde{X}_{f,i})$. This encourages invariance to the loss of localized spectral components and complements the dual spectral-temporal objectives. We adopt a lightweight tokenizer consisting of a temporal convolution, a spatial convolution, batch normalization, and pooling:

$$\begin{aligned} \tilde{z}_i &= \text{Tokenizer}(\tilde{x}_i) \\ &= \text{Pool}(\text{BatchNorm}(\text{Conv}_S(\text{Conv}_T(\tilde{x}_i)))) \end{aligned} \quad (1)$$

The resulting token embeddings $\tilde{z}_i$ lie in $\mathbb{R}^{N \times d}$, where N is the number of tokens and d the embedding dimension.

b) Dual temporal masking branches: EEG temporal dynamics reflect both causal progression, capturing directed event-related patterns such as onset-sustain-offset transitions in emotional responses, and contextual dependencies that encode longer-range temporal relationships such as emotion states. To jointly capture these complementary dependencies, STR employs two temporal branches: a bidirectional transformer that reconstructs masked tokens from surrounding context, and a causal transformer that predicts future representations from past context. A bidirectional transformer is trained with a random masking strategy, where a subset

of token positions $\mathcal{M}$ is replaced by mask tokens and the model reconstructs the corresponding input token $\tilde{z}_i$ from the unmasked context. Each reconstructed token is then mapped back to the input space through a two-layer MLP decoder:

$$g(\tilde{z}_i) = W_2 \sigma(W_1 \tilde{z}_i + b_1) + b_2, \quad (2)$$

where $\sigma(\cdot)$ denotes a non-linear activation, W_1 and W_2 are learnable weight matrices, and b_1 and b_2 are the corresponding bias terms. The reconstruction loss is computed against the original input segment:

$$\mathcal{L}_{ctx} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \|g(\tilde{z}_i) - x_i\|_2^2. \quad (3)$$

A causal transformer is optimized with a future mask, restricting each token at position i to attend only to $\{1, \dots, i\}$, thus preventing information leakage from the future. This imposes an autoregressive task in which the model predicts the next-token representation $\tilde{z}_{i+1}$. The prediction is decoded through the same two-layer MLP, yielding $g(\tilde{z}_{i+1})$, and the next-token loss is defined as

$$\mathcal{L}_{cau} = \frac{1}{N-1} \sum_{i=1}^{N-1} \|g(\tilde{z}_{i+1}) - x_{i+1}\|_2^2. \quad (4)$$

c) Joint spectral-temporal objective: The overall pre-training objective combines structural and temporal components:

$$\mathcal{L}_{pretrain} = \lambda_{ctx} \mathcal{L}_{ctx} + \lambda_{cau} \mathcal{L}_{cau}, \quad (5)$$

where λ_{ctx} and λ_{cau} are balancing coefficients (set to 1 by default). This design enforces that latent tokens must be decodable through $g(\cdot)$ back into the input domain, ensuring

that the learned representations remain both contextually and temporally consistent with the original signals.

B. EEG-Language Alignment with Multi-task Instruction Tuning

The goal of multi-task instruction tuning is to bridge the gap between EEG and language by learning conditionally aligned representations. To this end, we integrate semantic guidance from textual instructions into EEG embeddings and employ a compact set of latent queries to selectively attend to instruction-relevant neural patterns. The refined neural representation is then projected into a shared semantic space across tasks, where it is aligned with textual label prototypes to enable instruction-grounded decoding. Compared with coarse distribution-level alignment, which merely makes EEG and text statistically indistinguishable, our approach establishes explicit semantic correspondence that allows linguistic priors to directly shape and guide EEG representations.

a) Instruction-conditioned EEG-Language Interaction:

Previous EEG–language models such as NeuroLM achieve alignment through adversarial domain matching between neural and textual representations. In contrast, we proposed the Instruction-conditioned Q-Former (IQF). It achieves EEG–language alignment in a constructive and interpretable manner by conditioning neural representations on semantics and guiding cross-modal interaction. Let $\mathbf{m} \in \mathbb{R}^{2N \times d}$ denote the sequence of tokenized EEG embeddings obtained from the pretrained encoder, where the factor of 2 reflects the concatenation of the contextual and causal branch outputs, N is the number of tokens produced by each branch, and d is the EEG embedding dimension. Given the instruction text s_{ins} , we obtain its embedding $\mathbf{e}_{\text{ins}} \in \mathbb{R}^k$ using a frozen pretrained language encoder such as BERT [10] or SBERT [17]:

$$\mathbf{e}_{\text{ins}} = f_{\text{text}}(s_{\text{ins}}), \quad (6)$$

and we ℓ_2 -normalize the embedding:

$$\mathbf{e}_{\text{ins}} \leftarrow \frac{\mathbf{e}_{\text{ins}}}{\|\mathbf{e}_{\text{ins}}\|_2}. \quad (7)$$

This normalized vector serves as a high-level semantic prior to guide EEG representations toward the language space.

To fuse this conditioning prior with the EEG embedding space, we employ a Feature-wise Linear Modulation (FiLM) operator [18], which parameterizes an affine transformation. Specifically, the modulation parameters (γ, β) are derived from the instruction embedding via a nonlinear projection:

$$(\gamma, \beta) = \tanh(\mathbf{W}_{\gamma\beta} \mathbf{e}_{\text{ins}} + \mathbf{b}_{\gamma\beta}), \quad (8)$$

where $\mathbf{W}_{\gamma\beta} \in \mathbb{R}^{2d \times k}$ and $\mathbf{b}_{\gamma\beta} \in \mathbb{R}^{2d}$. We split (γ, β) into two d -dimensional vectors, $\gamma \in \mathbb{R}^d$ and $\beta \in \mathbb{R}^d$. For element-wise modulation, these vectors are broadcast from (d) to $(2N \times d)$ so that each latent dimension of every token is modulated by the same semantic prior. The instruction-conditioned representation is then obtained by

$$\tilde{\mathbf{m}} = (1 + \gamma) \odot \mathbf{m} + \beta, \quad (9)$$

where $\odot$ denotes element-wise multiplication. This formulation ensures that $\mathbf{m}$ is shaped by instruction semantics rather

than generic alignment. The instruction embedding $\mathbf{e}_{\text{ins}}$ biases the EEG latent space toward task-relevant features, producing representations that are both aligned with textual targets $\mathbf{e}_{\text{tgt}}$ and regularized on an instruction-informed manifold for improved semantic fidelity and generalization.

To extract instruction-relevant neural features from high-dimensional modulated EEG embeddings, we introduce a set of N_q learnable query vectors $\mathbf{Q}_0 \in \mathbb{R}^{N_q \times d}$, which function as compact latent probes. Rather than directly inheriting the full complexity of the EEG embedding space, these queries serve as bottlenecks through which information must be filtered. The proposed IQF employs cross-attention to couple $\mathbf{Q}_0$ with the instruction-modulated EEG embeddings $\tilde{\mathbf{m}}$, yielding

$$\begin{aligned} \mathbf{Q} &= \text{QFormer}(\mathbf{Q}_0, \tilde{\mathbf{m}}) \\ &= \text{softmax}\left(\frac{\mathbf{Q}_0 W_Q (\tilde{\mathbf{m}} W_K)^\top}{\sqrt{d}}\right) \tilde{\mathbf{m}} W_V, \end{aligned} \quad (10)$$

where W_Q, W_K, W_V denote the query, key, and value projections, and d is the key dimension. Through this operation, each query selectively attends to instruction-relevant EEG patterns encoded in $\tilde{\mathbf{m}}$, thereby performing instruction-guided feature extraction. Conceptually, this operation projects EEG embeddings onto a lower-dimensional query subspace regularized by the instruction prior. The learnable queries act as semantic filters, retaining task-relevant features while suppressing irrelevant variance, which yields embeddings that are more semantically consistent and generalizable. The resulting query outputs $\mathbf{Q}$ are aggregated and projected through a lightweight MLP to produce the instruction-aligned EEG representation $\hat{\mathbf{h}}$:

$$\hat{\mathbf{h}} = \frac{\mathbf{h}}{\|\mathbf{h}\|_2}, \quad \mathbf{h} = f_{\text{MLP}}(\text{vec}(\mathbf{Q})) \quad (11)$$

where $\|\cdot\|_2$ denotes the ℓ_2 (Euclidean) norm.

b) *Textual Prototype-based Semantic Alignment:* Given the ground-truth label $y \in \mathcal{C}$, we obtain its textual prototype embedding $\mathbf{e}_{\text{tgt}} \in \mathbb{R}^k$ by encoding the corresponding class name with the same frozen language model used for instructions:

$$\mathbf{e}_{\text{tgt}} = f_{\text{text}}(s_{\text{tgt}}), \quad (12)$$

Using the same encoder for both instructions and labels ensures that they are represented in a shared semantic space. To align the instruction-conditioned EEG representation with its semantic prototype, we minimize a cosine similarity loss:

$$\mathcal{L}_{\text{align}} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \left(1 - \cos(\hat{\mathbf{h}}, \mathbf{e}_{\text{tgt}}^c)\right) \mathbb{I}[y = c], \quad (13)$$

where $\mathbb{I}[y = c]$ is the indicator function for the ground-truth class. This objective encourages $\hat{\mathbf{h}}$ to be close to its corresponding prototype $\mathbf{e}_{\text{tgt}}$, thereby promoting class-discriminative organization in the shared EEG–language embedding space. As a result, EEG embeddings and language prototypes cohabit a shared latent space $\mathcal{Z}$, where distances reflect cross-modal semantic consistency and class-level discriminability.

C. Inference Procedure

To evaluate LEAF, we consider two inference regimes: (1) **Task-specific fine-tuning**: following multi-task instruction tuning, the model is further fine-tuned on the training split of each target dataset to adapt to its domain-specific distribution, while maintaining the same instruction–label formulation. This process does not introduce any additional classification head but refines the shared EEG–language parameters to better align with the target domain. (2) **Direct inference**: the model obtained after multi-task instruction tuning is kept fixed and directly evaluated on the test set of each downstream dataset. Given the corresponding textual instructions and target label embeddings $\mathbf{e}_{\text{tgt}}^y$, predictions are obtained through cosine similarity between $\hat{\mathbf{h}}$ and $\mathbf{e}_{\text{tgt}}^y$. In both regimes, the final prediction is computed as

$$\hat{y} = \arg \max_y \cos(\hat{\mathbf{h}}, \mathbf{e}_{\text{tgt}}^y),$$

where $\mathbf{e}_{\text{tgt}}^y$ is the embedding of the textual label for class y (e.g., “left hand”, “right hand”).

Notably, the language model used in LEAF is fully frozen, and all instruction and label embeddings are pre-computed offline, so it does not participate in either training or inference. Consequently, LEAF introduces no extra computational overhead from the text encoder, and the resulting model remains lightweight.

IV. EXPERIMENTS

A. Dataset

Pretraining datasets: We use 9 datasets for pretraining, namely Stieger2021 [19], SEED-FRA [20], SEED-GER [20], SEED-SD [21], SEED-Neg, ChineseEEG [22], Chisco [23], LargeSpanish [24], ThinkOutLoud [25] as the pretraining datasets. The total duration of these datasets is around 1153 hours. More details about the pretraining datasets can be found in the Appendix.

Instruct-tuning and downstream datasets: We systematically evaluate our LEAF on the five different BCI tasks with 16 datasets in total. *MI*: OpenBMI-MI [26], BCIC-IV-2a [27], BCIC-Upperlimb [28], SHU-MI [29], HighGamma [30], Cho2017 [31], Shin2017A [32], PhysioNet-MI [33]. *Emotion*: SEED [34], SEED-IV [35], SEED-V [36], SEED-VII [37], FACED [38]. *SSVEP*: OpenBMI-SSVEP [26]. *Covert speech*: BCIC2020-3 [28], *Mental Workload*: [39]. More details about the downstream datasets can be found in Appendix.

Table I summarises the train/validation/test split strategy adopted for each dataset. We employ cross-subject (CS) splits for datasets with sufficient participants, where held-out subjects are never seen during training, thereby directly testing generalisation across individuals. For the SEED-series emotion datasets and the covert-speech dataset, where the number of subjects is limited but multiple recording trials are available per subject, we instead adopt cross-trial (CT) splits. In all CS settings, 20% of the training subjects are reserved as a validation set for early stopping and hyperparameter selection. The number of classes ranges from 2 to 9, providing a diverse testbed for evaluating model robustness under varying task complexity.

B. Experimental Setup

a) **Baselines & Metrics**: In this paper, we selected both the state-of-the-art traditional models and the EEG-FMs as baselines. For the traditional models, we selected EEGNet [40], TSception [41], ST-Transformer [42] and Conformer [43]. For the EEG foundation model, we selected BIOT [15], EEGPT [2], LaBraM [3], and CBraMod [4]. Implementation about the baselines can be found in Appendix E. To provide a reliable evaluation across imbalanced datasets, we adopted **balanced accuracy** and **Cohen’s Kappa** as performance metrics. Balanced accuracy accounts for class imbalance by averaging recall across classes, while Cohen’s Kappa measures the agreement between predicted and true labels beyond chance level, providing a more robust assessment of model performance.

b) **EEG Preprocessing and Unification**: EEG recordings from different studies typically use diverse electrode montages. As illustrated in Fig. 8, LEAF performs channel unification by interpolating all signals onto the standardized 10–10 electrode layout with 65 channels. For datasets recorded with fewer than 65 channels, spatial interpolation is applied to enforce a consistent topological structure across inputs, for datasets with more than 65 channels, the 65 channels corresponding to the 10–10 system are selected. Signals are then downsampled to 200 Hz. All MI datasets are band-pass filtered to 0.3–40 Hz, while all other datasets are filtered to 0.3–70 Hz. For segmentation, we distinguish between pre-segmented and continuous datasets. For pre-segmented datasets such as MI, SSVEP, and Covert Speech, we directly use the original trial-based divisions provided by each dataset. For continuous datasets, we follow dataset-specific conventions: FACED recordings are divided into 10-second windows, the SEED series datasets are split into 4-second segments, and the Workload dataset is segmented into 5-second windows.

c) **Implementation Details**: Pre-training and instruction tuning are both conducted in an end-to-end manner. Detailed model hyperparameters and training parameters are provided in Appendix A. Our model is trained on a 4×H100 GPU cluster using PyTorch, with a total of approximately 26.42 M parameters—substantially smaller than most competing EEG foundation models.

C. Experimental Results

a) **Task-specific Finetuning**: In the task-specific finetuning setting, LEAF is first trained with multi-task instruction tuning on the combined training split of all datasets. It is then further finetuned using only the training set of each target downstream dataset. This setup ensures a fair comparison against all baselines including EEGNet, Conformer, TSception, ST-Transformer, BIOT, EEGPT, LaBraM, and CBraMod. All averaged accuracies are reported over three random seeds in Table III.

Motor Imagery: LEAF performs particularly strongly on MI benchmarks, achieving the best results on five datasets, including BCIC-IV-2a (63.81%), HighGamma (79.82%), Cho2017 (79.08%), and Shin2017A (72.56%). These results highlight robust generalization across subjects and recording

TABLE I: Train/validation/test split strategies for downstream EEG datasets. CS = Cross-Subject; CT = Cross-Trial.

#	Dataset	Split	Train	Validation	Test	(#cls) Classes
<i>Motor Imagery</i>						
1	BCIC-IV-2a	CS	Subj. 1–7	20% of train	Subj. 8–9	(4) Left, Right, Foot, Tongue
2	OpenBMI-MI	CS	Subj. 1–42	20% of train	Subj. 43–54	(2) Right, Left
3	BCIC-Upperlimb	CS	Subj. 1–11	20% of train	Subj. 12–15	(3) Cylin, Sphe, Lumbrical
4	SHU-MI	CS	Subj. 1–20	20% of train	Subj. 21–25	(2) Right, Left
5	HighGamma	CS	Subj. 1–10	20% of train	Subj. 11–14	(3) Left, Right, Foot
6	Cho2017	CS	Subj. 1–40	20% of train	Subj. 41–49	(2) Left, Right
7	Shin2017A	CS	Subj. 1–22	20% of train	Subj. 23–28	(2) Left, Right
8	PhysioNet-MI	CS	Subj. 1–80	20% of train	Subj. 81–109	(2) Left, Right
<i>Emotion Recognition</i>						
9	FACED	CS	Subj. 1–100	20% of train	Subj. 101–122	(9) Anger, Fear, Disgust, Sad, Amusement, Inspiration, ...
10	SEED	CT	Trial 1–9	Trial 10–12	Trial 13–15	(3) Positive, Neutral, Negative
11	SEED-IV	CT	Trial 1–16	Trial 17–20	Trial 21–24	(4) Neutral, Sad, Fear, Happy
12	SEED-V	CT	Trial 1–5	Trial 6–10	Trial 11–15	(5) Disgust, Fear, Sad, Neutral, Happy
13	SEED-VII	CT	Trial 1–10	Trial 11–15	Trial 16–20	(7) Happy, Surprise, Neutral, Sad, Disgust, Fear, Anger
<i>SSVEP, Workload, Covert Speech</i>						
14	OpenBMI-SSVEP	CS	Subj. 1–42	20% of train	Subj. 43–54	(4) 12.0, 8.6, 6.6, 5.4 Hz
15	Mental Workload	CS	Subj. 0–31	20% of train	Subj. 32–35	(2) Resting, Workload
16	BCIC-Speech	CT	Trial 1–250	Trial 251–300	Trial 301–350	(5) hello, help-me, stop, thank-you, yes

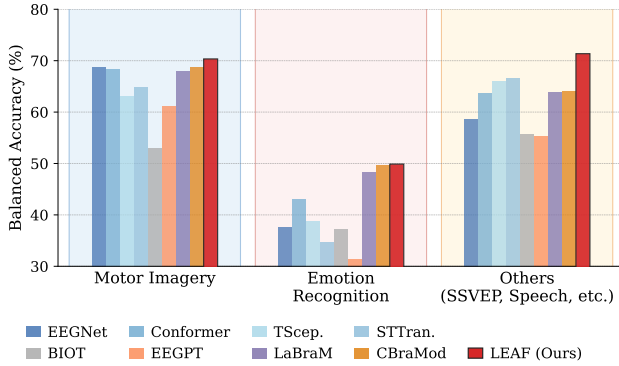


Fig. 2: Average balanced accuracy (%) per paradigm group for all models. LEAF consistently achieves the highest accuracy across motor imagery, emotion recognition, and the remaining paradigms (SSVEP, covert speech, mental workload).

setups. **Emotion Recognition:** LEAF also establishes the strongest performance on four emotion datasets, with especially clear gains on FACED (58.19%), SEED-IV (46.3%), SEED-V (41.26%), and SEED-VII (33.56%). Together with the strong MI results above, this trend suggests that language-guided alignment is particularly beneficial for motor imagery and affective decoding tasks, where task instructions and label semantics provide informative priors for discrimination. **Other paradigms:** Beyond MI and emotion recognition, LEAF remains competitive across the remaining paradigms, achieving the best performance on OpenBMI-SSVEP (94.62%), BCIC-Speech (54.53%), and Mental Workload (64.93%). Overall, LEAF delivers the strongest average performance across paradigm groups, substantially surpassing all baselines and demonstrating strong transferability across diverse EEG decoding scenarios. Figure 2 further summarizes the balanced-accuracy comparison at the paradigm level, and detailed results with standard deviations are provided in the appendix.

Table II summarises the paradigm-level averages. The upper

TABLE II: Average balanced accuracy (%) and Cohen’s Kappa (%) per paradigm group.

Method	MI		Emotion		Other		Overall	
	B.Acc	Kappa	B.Acc	Kappa	B.Acc	Kappa	B.Acc	Kappa
EEGNet	68.74	44.41	37.56	21.56	58.70	37.05	57.11	35.89
Conformer	68.45	42.45	43.12	28.28	63.78	43.86	59.66	38.29
TScep.	63.14	34.60	38.85	23.50	66.07	48.58	56.10	33.75
STTran.	64.79	38.38	34.80	18.29	66.66	49.03	55.77	34.10
BIOT	53.08	19.71	37.33	21.73	55.71	32.92	48.65	22.82
EEGPT	61.23	33.50	31.33	13.63	55.30	32.66	50.77	27.14
LaBraM	68.00	42.91	48.42	35.41	63.89	45.92	61.11	41.13
CBraMod	68.87	43.08	49.67	36.68	64.10	44.93	61.97	41.43
LEAF	70.34	46.06	49.88	36.78	71.36	55.78	64.14	44.98

block lists task-specific architectures (EEGNet, Conformer, TSception, ST-Transformer), while the lower block contains EEG foundation models (BIOT, EEGPT, LaBraM, CBraMod). LEAF outperforms all baselines across every paradigm group and metric. The largest gains appear in the “Other” category (SSVEP, covert speech, mental workload), where LEAF achieves 71.36% balanced accuracy and 55.78% Kappa—surpassing the next-best method by over 4.7 and 6.8 percentage points, respectively. These results confirm that LEAF generalises effectively not only within well-studied paradigms such as MI and emotion recognition, but also to less common BCI tasks.

b) *Direct Inference:* In the direct inference setting, LEAF is trained on the combined training split of all datasets during the multi-task instruction tuning stage and then directly evaluated on each downstream dataset without any additional finetuning. Since the instruction condition influences direct inference performance, we report results under three instruction settings: No-instruction: input *Default* or *None*; Task-only instruction: provide an instruction specifying the EEG task; Task & Target Instruction: provide both the task type and target classes (e.g., This is an MI task; decode *Left* vs. *Right*). Figure 3 presents the per-dataset balanced accuracy obtained under these three instruction settings. A clear overall

TABLE III: Performance comparison on 16 datasets (Balanced Accuracy / Cohen’s Kappa), reported as mean $\pm$ std over three random seeds.

(a) Motor Imagery (Part I)								
Methods	BCIC-IV-2a		OpenBMI-MI		BCIC-Upperlimb		SHU-MI	
	B.Acc	Kappa	B.Acc	Kappa	B.Acc	Kappa	B.Acc	Kappa
EEGNet	0.6369 $\pm$ 0.0097	0.4685 $\pm$ 0.0130	0.8170 $\pm$ 0.0032	0.6259 $\pm$ 0.0114	0.5281 $\pm$ 0.0046	0.2962 $\pm$ 0.0062	0.5665 $\pm$ 0.0007	0.1976 $\pm$ 0.0015
Conformer	0.6347 $\pm$ 0.0118	0.4663 $\pm$ 0.0157	0.8274$\pm$0.0027	0.6387$\pm$0.0098	0.5473 $\pm$ 0.0039	0.3120 $\pm$ 0.0053	0.5780 $\pm$ 0.0008	0.1591 $\pm$ 0.0018
TScep.	0.6252 $\pm$ 0.0106	0.4543 $\pm$ 0.0141	0.6665 $\pm$ 0.0032	0.3425 $\pm$ 0.0116	0.5199 $\pm$ 0.0046	0.2792 $\pm$ 0.0062	0.5564 $\pm$ 0.0007	0.2047 $\pm$ 0.0015
STTran.	0.5816 $\pm$ 0.0100	0.4334 $\pm$ 0.0133	0.7514 $\pm$ 0.0030	0.5046 $\pm$ 0.0107	0.5284 $\pm$ 0.0037	0.2972 $\pm$ 0.0050	0.5714 $\pm$ 0.0008	0.2191 $\pm$ 0.0018
BIOT	0.5139 $\pm$ 0.0099	0.3898 $\pm$ 0.0132	0.5613 $\pm$ 0.0029	0.1225 $\pm$ 0.0105	0.4595 $\pm$ 0.0046	0.1843 $\pm$ 0.0063	0.5587 $\pm$ 0.0008	0.2151 $\pm$ 0.0016
EEGPT	0.5279 $\pm$ 0.0123	0.4617 $\pm$ 0.0164	0.7323 $\pm$ 0.0026	0.4624 $\pm$ 0.0093	0.5231 $\pm$ 0.0044	0.2031 $\pm$ 0.0060	0.5228 $\pm$ 0.0009	0.1657 $\pm$ 0.0019
LaBraM	0.6234 $\pm$ 0.0125	0.4599 $\pm$ 0.0166	0.7874 $\pm$ 0.0032	0.5765 $\pm$ 0.0116	0.5414 $\pm$ 0.0037	0.3108 $\pm$ 0.0050	0.6338 $\pm$ 0.0008	0.2338 $\pm$ 0.0016
CBraMod	0.6139 $\pm$ 0.0117	0.4385 $\pm$ 0.0156	0.7895 $\pm$ 0.0028	0.5828 $\pm$ 0.0101	0.5454 $\pm$ 0.0042	0.3145 $\pm$ 0.0057	0.6403$\pm$0.0007	0.2384$\pm$0.0015
LEAF	0.6381$\pm$0.0120	0.4692$\pm$0.0160	0.8134 $\pm$ 0.0031	0.6127 $\pm$ 0.0112	0.5483$\pm$0.0038	0.3187$\pm$0.0057	0.6134 $\pm$ 0.0008	0.2267 $\pm$ 0.0017
(b) Motor Imagery (Part II)								
Methods	HighGamma		Cho2017		Shin2017A		PhysioNet-MI	
	B.Acc	Kappa	B.Acc	Kappa	B.Acc	Kappa	B.Acc	Kappa
EEGNet	0.7856 $\pm$ 0.0038	0.5838 $\pm$ 0.0076	0.7644 $\pm$ 0.0062	0.5310 $\pm$ 0.0124	0.7054 $\pm$ 0.0085	0.4521 $\pm$ 0.0171	0.6953 $\pm$ 0.0029	0.3980 $\pm$ 0.0177
Conformer	0.7637 $\pm$ 0.0036	0.5627 $\pm$ 0.0071	0.7838 $\pm$ 0.0057	0.5719 $\pm$ 0.0115	0.6464 $\pm$ 0.0071	0.2901 $\pm$ 0.0142	0.6951 $\pm$ 0.0034	0.3952 $\pm$ 0.0208
TScep.	0.6851 $\pm$ 0.0031	0.5129 $\pm$ 0.0061	0.7339 $\pm$ 0.0056	0.4640 $\pm$ 0.0112	0.5992 $\pm$ 0.0089	0.1891 $\pm$ 0.0179	0.6649 $\pm$ 0.0030	0.3216 $\pm$ 0.0182
STTran.	0.7019 $\pm$ 0.0028	0.5162 $\pm$ 0.0056	0.7612 $\pm$ 0.0059	0.5260 $\pm$ 0.0117	0.6168 $\pm$ 0.0093	0.2371 $\pm$ 0.0186	0.6706 $\pm$ 0.0035	0.3370 $\pm$ 0.0217
BIOT	0.5824 $\pm$ 0.0035	0.4855 $\pm$ 0.0069	0.5413 $\pm$ 0.0055	0.0729 $\pm$ 0.0110	0.5422 $\pm$ 0.0094	0.0909 $\pm$ 0.0188	0.4874 $\pm$ 0.0030	0.0162 $\pm$ 0.0184
EEGPT	0.6516 $\pm$ 0.0038	0.5152 $\pm$ 0.0075	0.7197 $\pm$ 0.0058	0.4305 $\pm$ 0.0115	0.5389 $\pm$ 0.0074	0.0688 $\pm$ 0.0148	0.6820 $\pm$ 0.0029	0.3730 $\pm$ 0.0180
LaBraM	0.6939 $\pm$ 0.0036	0.5200 $\pm$ 0.0072	0.7614 $\pm$ 0.0053	0.5221 $\pm$ 0.0106	0.6744 $\pm$ 0.0073	0.3607 $\pm$ 0.0145	0.7246$\pm$0.0028	0.4487$\pm$0.0174
CBraMod	0.7684 $\pm$ 0.0034	0.5671 $\pm$ 0.0068	0.7439 $\pm$ 0.0059	0.4974 $\pm$ 0.0117	0.6861 $\pm$ 0.0072	0.3689 $\pm$ 0.0144	0.7219 $\pm$ 0.0037	0.4386 $\pm$ 0.0229
LEAF	0.7982$\pm$0.0035	0.5965$\pm$0.0070	0.7908$\pm$0.0060	0.5816$\pm$0.0120	0.7256$\pm$0.0088	0.4811$\pm$0.0176	0.6992 $\pm$ 0.0035	0.3983 $\pm$ 0.0214
(c) Emotion Recognition								
Methods	FACED		SEED		SEED-IV		SEED-V	
	B.Acc	Kappa	B.Acc	Kappa	B.Acc	Kappa	B.Acc	Kappa
EEGNet	0.4271 $\pm$ 0.0019	0.3512 $\pm$ 0.0357	0.5337 $\pm$ 0.0122	0.3143 $\pm$ 0.0251	0.3651 $\pm$ 0.0102	0.1578 $\pm$ 0.0143	0.2932 $\pm$ 0.0102	0.1136 $\pm$ 0.0143
Conformer	0.4943 $\pm$ 0.0019	0.4263 $\pm$ 0.0350	0.6254 $\pm$ 0.0125	0.4342 $\pm$ 0.0186	0.4094 $\pm$ 0.0112	0.2121 $\pm$ 0.0174	0.3060 $\pm$ 0.0112	0.1335 $\pm$ 0.0174
TScep.	0.2056 $\pm$ 0.0022	0.1088 $\pm$ 0.0410	0.6369 $\pm$ 0.0155	0.4604 $\pm$ 0.0289	0.4063 $\pm$ 0.0078	0.1876 $\pm$ 0.0139	0.3637 $\pm$ 0.0078	0.1984 $\pm$ 0.0139
STTran.	0.3791 $\pm$ 0.0023	0.2999 $\pm$ 0.0438	0.5882 $\pm$ 0.0079	0.3873 $\pm$ 0.0133	0.3616 $\pm$ 0.0072	0.1415 $\pm$ 0.0121	0.2244 $\pm$ 0.0072	0.0344 $\pm$ 0.0121
BIOT	0.1711 $\pm$ 0.0022	0.0647 $\pm$ 0.0407	0.6674 $\pm$ 0.0118	0.5034 $\pm$ 0.0254	0.4141 $\pm$ 0.0187	0.1937 $\pm$ 0.0262	0.3045 $\pm$ 0.0187	0.1306 $\pm$ 0.0262
EEGPT	0.3346 $\pm$ 0.0025	0.2486 $\pm$ 0.0469	0.5054 $\pm$ 0.0105	0.2659 $\pm$ 0.0191	0.3202 $\pm$ 0.0144	0.0861 $\pm$ 0.0210	0.2253 $\pm$ 0.0144	0.0335 $\pm$ 0.0210
LaBraM	0.5457 $\pm$ 0.0021	0.4809 $\pm$ 0.0393	0.7083 $\pm$ 0.0107	0.5613 $\pm$ 0.0188	0.4415 $\pm$ 0.0138	0.2560 $\pm$ 0.0209	0.4010 $\pm$ 0.0138	0.2563 $\pm$ 0.0209
CBraMod	0.5787 $\pm$ 0.0021	0.4941 $\pm$ 0.0385	0.7102$\pm$0.0089	0.5868$\pm$0.0122	0.4605 $\pm$ 0.0097	0.2728 $\pm$ 0.0143	0.4029 $\pm$ 0.0138	0.2570 $\pm$ 0.0209
LEAF	0.5819$\pm$0.0023	0.5243$\pm$0.0430	0.7011 $\pm$ 0.0068	0.5543 $\pm$ 0.0176	0.4630$\pm$0.0034	0.2754$\pm$0.0214	0.4126$\pm$0.0023	0.2575$\pm$0.0430
(d) Emotion (SEED-VII), SSVEP, Covert Speech, and Workload								
Methods	SEED-VII		OpenBMI-SSVEP		BCIC-Speech		Mental Workload	
	B.Acc	Kappa	B.Acc	Kappa	B.Acc	Kappa	B.Acc	Kappa
EEGNet	0.2587 $\pm$ 0.0010	0.1413 $\pm$ 0.0345	0.9430 $\pm$ 0.0036	0.9253 $\pm$ 0.0265	0.2699 $\pm$ 0.0023	0.0880 $\pm$ 0.0413	0.5480 $\pm$ 0.0224	0.0982 $\pm$ 0.0170
Conformer	0.3209 $\pm$ 0.0012	0.2081 $\pm$ 0.0439	0.8981 $\pm$ 0.0045	0.8634 $\pm$ 0.0337	0.4170 $\pm$ 0.0019	0.2722 $\pm$ 0.0330	0.5984 $\pm$ 0.0185	0.1801 $\pm$ 0.0140
TScep.	0.3300 $\pm$ 0.0011	0.2198 $\pm$ 0.0394	0.8026 $\pm$ 0.0043	0.7375 $\pm$ 0.0322	0.5314 $\pm$ 0.0024	0.4164 $\pm$ 0.0426	0.6480 $\pm$ 0.0177	0.3034 $\pm$ 0.0134
STTran.	0.1867 $\pm$ 0.0011	0.0516 $\pm$ 0.0400	0.9398 $\pm$ 0.0046	0.9195 $\pm$ 0.0341	0.4266 $\pm$ 0.0019	0.2832 $\pm$ 0.0347	0.6335 $\pm$ 0.0179	0.2683 $\pm$ 0.0136
BIOT	0.3096 $\pm$ 0.0011	0.1939 $\pm$ 0.0402	0.7936 $\pm$ 0.0040	0.7258 $\pm$ 0.0297	0.2920 $\pm$ 0.0024	0.1138 $\pm$ 0.0427	0.5857 $\pm$ 0.0222	0.1480 $\pm$ 0.0168
EEGPT	0.1809 $\pm$ 0.0013	0.0476 $\pm$ 0.0470	0.9301 $\pm$ 0.0047	0.9098 $\pm$ 0.0351	0.2364 $\pm$ 0.0022	0.0478 $\pm$ 0.0397	0.4925 $\pm$ 0.0192	0.0223 $\pm$ 0.0145
LaBraM	0.3244 $\pm$ 0.0011	0.2159 $\pm$ 0.0382	0.8698 $\pm$ 0.0041	0.8269 $\pm$ 0.0302	0.4819 $\pm$ 0.0024	0.3863 $\pm$ 0.0434	0.5650 $\pm$ 0.0181	0.1643 $\pm$ 0.0137
CBraMod	0.3311 $\pm$ 0.0010	0.2233 $\pm$ 0.0362	0.9205 $\pm$ 0.0039	0.8924 $\pm$ 0.0291	0.4280 $\pm$ 0.0022	0.2860 $\pm$ 0.0395	0.5746 $\pm$ 0.0200	0.1695 $\pm$ 0.0151
LEAF	0.3356$\pm$0.0012	0.2275$\pm$0.0430	0.9462$\pm$0.0043	0.9283$\pm$0.0320	0.5453$\pm$0.0023	0.4317$\pm$0.0410	0.6493$\pm$0.0211	0.3134$\pm$0.0160

trend emerges: as the instruction becomes more informative, direct-inference performance improves on most datasets. This result suggests that task and target semantics act as effective priors, guiding EEG embeddings toward more discriminative and semantically aligned representations.

D. Analysis of Instruction-Guided EEG Representations

To comprehensively understand how instructions shape the latent EEG representation space, we present both qualitative visualizations and quantitative metrics. Figure 4 shows a UMAP [44] projection of the learned embeddings, where kernel density estimation (KDE) highlights the concentration of samples. Class labels are represented by fixed textual prototypes that are projected through the same model and UMAP mapping. Without instructions, the embeddings exhibit

weak structure with substantial overlap between classes. When instructions are provided, however, the feature space becomes more organized: class clusters become clearly separated and better aligned with their corresponding semantic prototypes.

To complement the qualitative visualization, we further quantify the effect of instruction conditioning using two embedding-level metrics: intra-class variance, which measures class-wise compactness, and inter-class distance, which captures the separation between class centers. For class c with embedding vectors $\{z_i\}_{i \in C_c}$ and class mean μ_c , these metrics are defined as

$$\text{Intra} = \frac{1}{K} \sum_{c=1}^K \frac{1}{|C_c|} \sum_{i \in C_c} \|z_i - \mu_c\|_2^2, \quad (14)$$

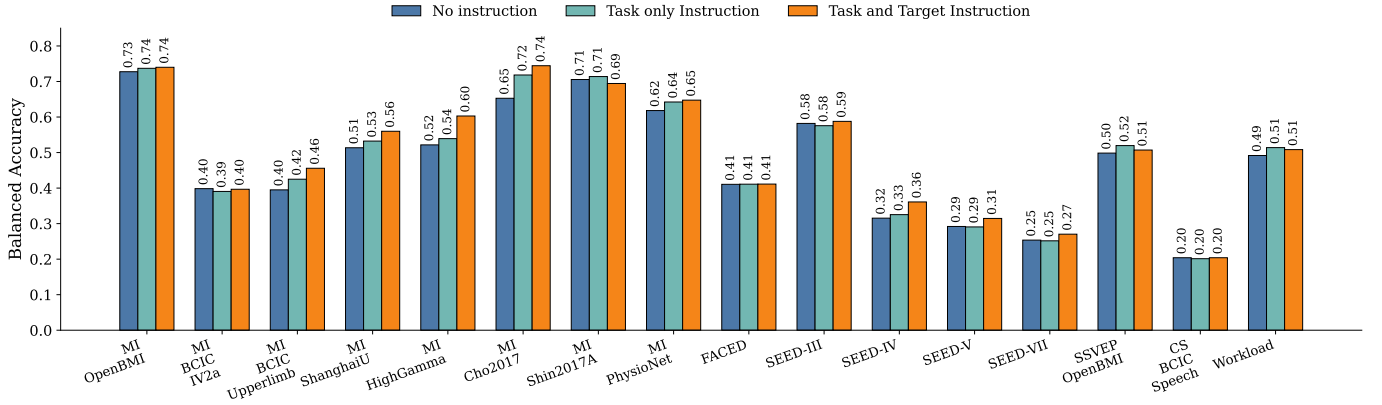


Fig. 3: Per-dataset balanced accuracy under three instruction conditions in direct inference: no instruction, task-only instruction, and task-plus-target instruction. Across most datasets, richer instructions yield consistent gains, with the strongest improvements appearing in several motor imagery and emotion-recognition benchmarks.

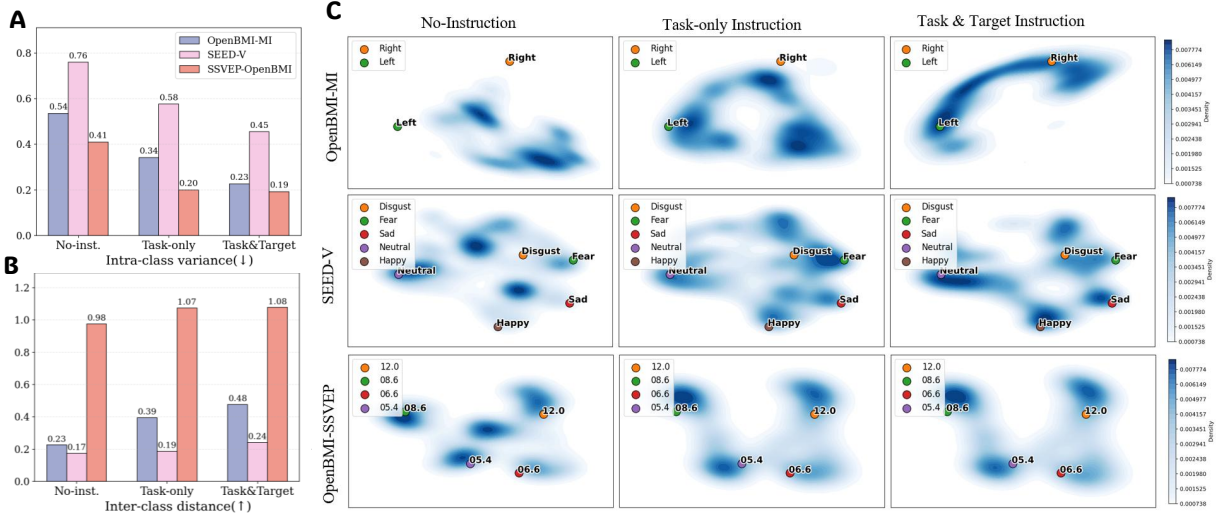


Fig. 4: Effect of instruction conditioning on EEG representations. A: intra-class variance. B: inter-class distance. Richer instructions reduce intra-class variance and increase inter-class distance. C: KDE-based embedding distributions. Without instructions, class-conditional embeddings overlap substantially, whereas task and target instructions yield more compact, better separated clusters aligned with their textual prototypes.

$$\text{Inter} = \frac{2}{K(K-1)} \sum_{c < d} \|\mu_c - \mu_d\|_2. \quad (15)$$

where K is the number of classes. Smaller intra-class variance indicates tighter clusters and stronger within-class consistency, whereas larger inter-class distance reflects clearer class separation and stronger discriminative structure.

As shown in Fig. 4, instruction-conditioned models consistently exhibit lower intra-class variance and larger inter-class distances. Together with the embedding visualization, these results demonstrate that natural-language instructions act as explicit semantic constraints that reshape the EEG embedding space toward a more discriminative and linguistically grounded organization.

Beyond the embedding-space analysis above, we further examine how LEAF captures task-dependent spatial EEG patterns across representative downstream datasets. Figure 7

presents the corresponding saliency topographies, offering a complementary view of the model’s learned representations.

V. ABLATION STUDY

A. Effect of Joint Spectral-Temporal Reconstruction

To evaluate the contribution of each masking strategy during STR pretraining, we performed an ablation study in which individual components were selectively removed. Results in Table IV indicate that across OpenBMI-MI, BCI-IV-2a, and SEED, retaining all three masking strategies yields the best B-Acc/Kappa. These results confirm that frequency masking, random temporal masking, and causal masking each capture complementary aspects of EEG structure, and their combination produces the most generalizable representations.

To better understand the role of the learnable queries in IQF, Fig. 5 visualizes their layer-wise evolution using PCA on four

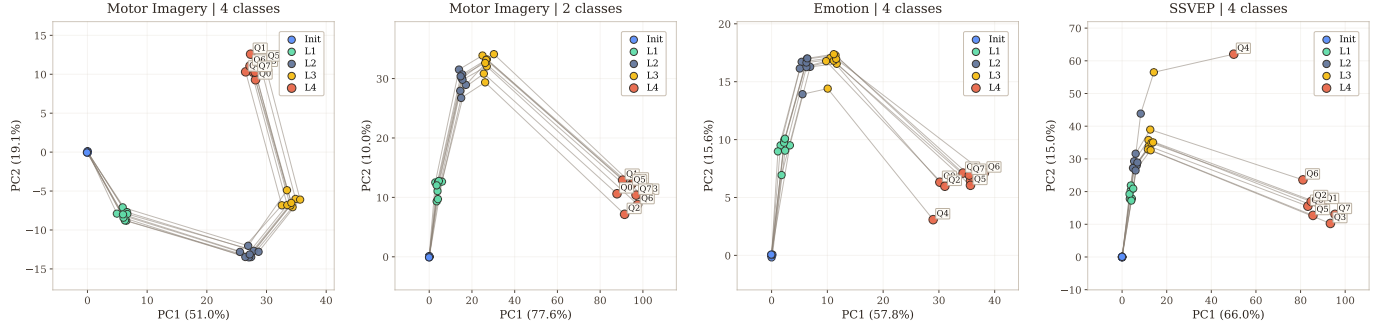


Fig. 5: PCA visualization of Q-Former query evolution on four representative EEG decoding tasks. Queries diverge from their shared initialization and gradually form task-specific configurations across layers.

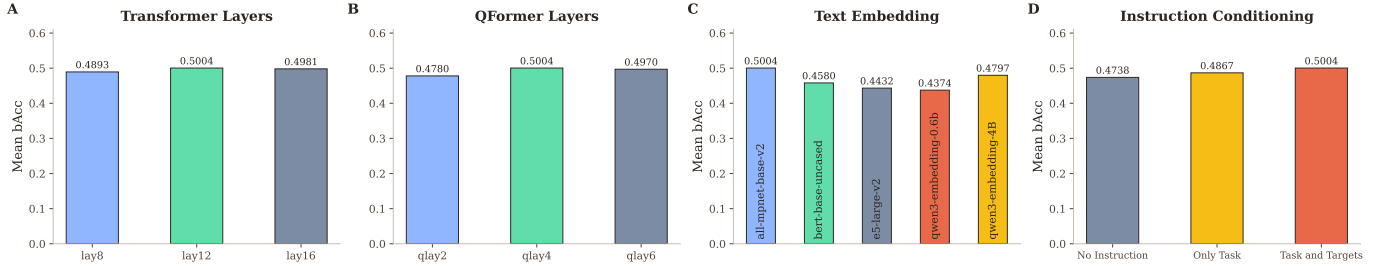


Fig. 6: Comprehensive ablation summary of LEAF. (A) Transformer depth ablation under Task & Target Instruction comparing 8, 12, and 16 layers. (B) Q-Former depth ablation under Task & Target Instruction comparing 2, 4, and 6 layers. (C) Text encoder ablation under Task & Target Instruction comparing embedding models. (D) Instruction conditioning ablation comparing No instruction, Task-only instruction, and Task & Target Instruction. Values denote mean balanced accuracy.

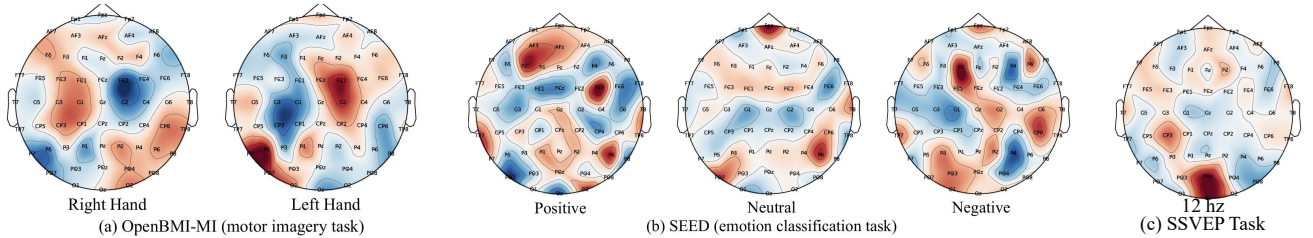


Fig. 7: Topography visualization on downstream datasets, illustrating the spatial EEG activation patterns captured by LEAF across different BCI task categories.

TABLE IV: Ablation study of masking strategies. Best results are highlighted in bold.

Freq.	Rand.	Causal	OpenBMI-MI	SEED
✓			0.8078 / 0.6195	0.6882 / 0.5487
✓	✓		0.7896 / 0.6017	0.6771 / 0.5298
✓	✓	✓	0.8033 / 0.6152	0.6957 / 0.5499
✓	✓	✓	0.8144 / 0.6287	0.7011 / 0.5543

representative EEG decoding tasks: 4-class motor imagery, 2-class motor imagery, 4-class emotion recognition, and 4-class SSVEP. Starting from a shared initialization, the queries progressively diverge as they pass through the Q-Former and eventually organize into task-specific configurations. This observation suggests that the latent query set gradually specializes to capture task-relevant semantic structure, which helps explain why strong performance can be achieved with only a

small number of learnable queries.

To provide a consolidated view of the main design choices, Fig. 6 summarizes the average balanced accuracy across four ablation axes. The architecture ablations in Fig. 6(A)–(B) show that performance is strongest with a 12-layer EEG transformer and a 4-layer Q-Former, indicating that this configuration provides the most favorable balance between model capacity and transferability. Figure 6(C) further shows that all-mpnet-base-v2 is the most effective frozen text encoder for LEAF, outperforming BERT, E5, and both Qwen3 embedding variants in this alignment setting. Finally, Fig. 6(D) shows that richer instruction conditioning yields a monotonic improvement in mean performance: the average balanced accuracy increases from 0.4738 without instructions to 0.4867 with task-level instructions and further to 0.5004 when both task and target semantics are provided. Taken together, these ablations show that LEAF does not rely on a single dominant

design choice; rather, its gains arise from the complementary interaction between a moderately sized EEG backbone, a sufficiently expressive query transformer, a strong frozen semantic encoder, and informative instruction prompts. This overall consistency also suggests that the final model configuration is not merely tuned to one dataset or one paradigm, but captures a broadly transferable alignment strategy for EEG decoding. These trends are consistent with the detailed analyses above and support the final LEAF configuration used in all main experiments.

VI. CONCLUSION

We introduced LEAF, a foundation model for EEG–Language Alignment with Semantic Task Instruction and Querying. By combining a joint Spectral-Temporal Reconstruction module and an Instruction-conditioned Q-Former, LEAF learns language-guided EEG representations that transfer effectively across tasks. Extensive evaluations on 16 downstream datasets covering MI, emotion recognition, SSVEP, covert speech, and healthcare applications show that LEAF achieves, on average, state-of-the-art (SOTA) performance, highlighting the value of instruction-informed alignment for generalizable EEG decoding. Our results further suggest that natural language can serve as both an interpretable semantic anchor and a transferable source of supervision for future EEG foundation models and BCI systems.

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APPENDIX A PARAMETER SETTINGS

Table V summarises the model architecture, training configuration, and parameter breakdown of LEAF.

TABLE V: Architecture, training, and parameter settings of LEAF.

Setting	Value
<i>Tokenization</i>	
Sampling rate	200 Hz
Segment window length	0.5 s
Input channels	65
Temporal kernel size	(1, 40), padding (1, 20)
Spatial kernel size	(65, 1)
Pooling	(1, 10)
<i>Transformer Encoder</i>	
Layers	12
Token size	256
Feed-forward scale	4
Attention heads	8
Dropout	0.1
Mask ratio	0.5
Frequency cutoff range	1–50 Hz
Frequency cutoff band	6 Hz continuous band
<i>Q-Former</i>	
Number of queries	8
Layers	4
Text embedding dim.	768
Text encoder	all-mpnet-base-v2
<i>Training</i>	
Batch size	512
Peak learning rate	1×10^{-3}
LR scale (transformer)	0.1
LR scale (other)	1.0
Min. learning rate	1×10^{-4}
LR scheduler	Cosine annealing
Optimizer	AdamW ($\beta_{1,2} = 0.9, 0.999$)
Weight decay	1×10^{-3}
Precision	bf16-mixed
<i>Parameter Count</i>	
Tokenizer	0.27 M
Dual Transformer	20.23 M
Q-Former	6.19 M
Total	26.42 M

APPENDIX B EFFECT OF INCORRECT INSTRUCTIONS

To further examine the role of language guidance, we analyze cases where the model is deliberately given misleading instructions that do not match the underlying EEG dataset. Figure 9 shows examples on OpenBMI-MI and SEED-V datasets.

When provided with correct instructions, the learned feature spaces become more structured, with compact intra-class clusters and clearer inter-class separation. However, when misleading instructions are introduced, the feature space is distorted toward the semantics of the given instruction rather than the ground-truth task. For example, MI data conditioned on emotion-related instructions form clusters resembling affective categories, and SEED-V data prompted with MI or SSVEP instructions are reorganized into motor or frequency-based groupings. These results emphasize the strong controllability of our model through natural language. While correct

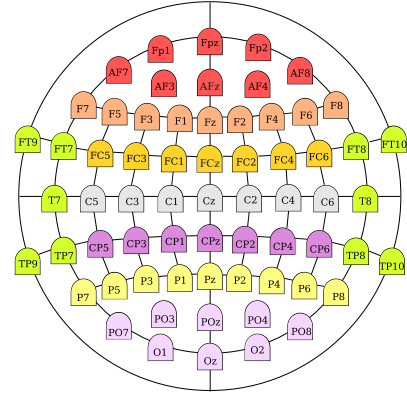


Fig. 8: Standardized 65-channel EEG montage used by LEAF. Inputs from different datasets are spatially interpolated to this common 10–10 electrode configuration before being processed by the model.

instructions enhance discriminability, misleading instructions actively reshape the representation space according to the semantic prior they provide. This highlights both the power and sensitivity of instruction-conditioned alignment in EEG-FMs.

APPENDIX C DETAILS ABOUT PRE-TRAINING DATASETS

- Stieger2021 [19]: This database contains EEG recordings from 62 healthy participants, each completing 7–11 sessions of BCI training to control a computer cursor in one- and two-dimensional spaces using motor imagery. Data were collected with 62 electrodes, and accompanying behavioral measures.
- SEED-FRA [20]: Eight French-speaking subjects participated in an emotion elicitation paradigm using 21 film clips covering positive, neutral, and negative emotional categories. EEG was recorded with the same 62-channel NeuroScan setup as SEED, enabling cross-lingual analysis of affective neural responses.
- SEED-GER [20]: Eight German-speaking subjects participated under the same protocol as SEED-FRA, using 20 film clips spanning positive, neutral, and negative emotions. Together with SEED-FRA, this dataset supports investigation of language- and culture-related variations in EEG-based emotion recognition.
- SEED-SD [21]: SEED-SD is a multimodal EEG and eye-tracking dataset collected from 40 healthy participants under three sleep-related conditions—sleep deprivation, sleep recovery, and normal sleep. In each condition, participants watched 24 video clips (six per emotion) designed to evoke four basic emotions: happiness, sadness, fear, and neutral; each clip lasts about 2.5 minutes.
- ChineseEEG [22]: This dataset consists of EEG data recorded from 10 participants silently reading approximately 13 hours of Chinese text from two well-known novels, providing a large-scale corpus for language-related neural decoding.

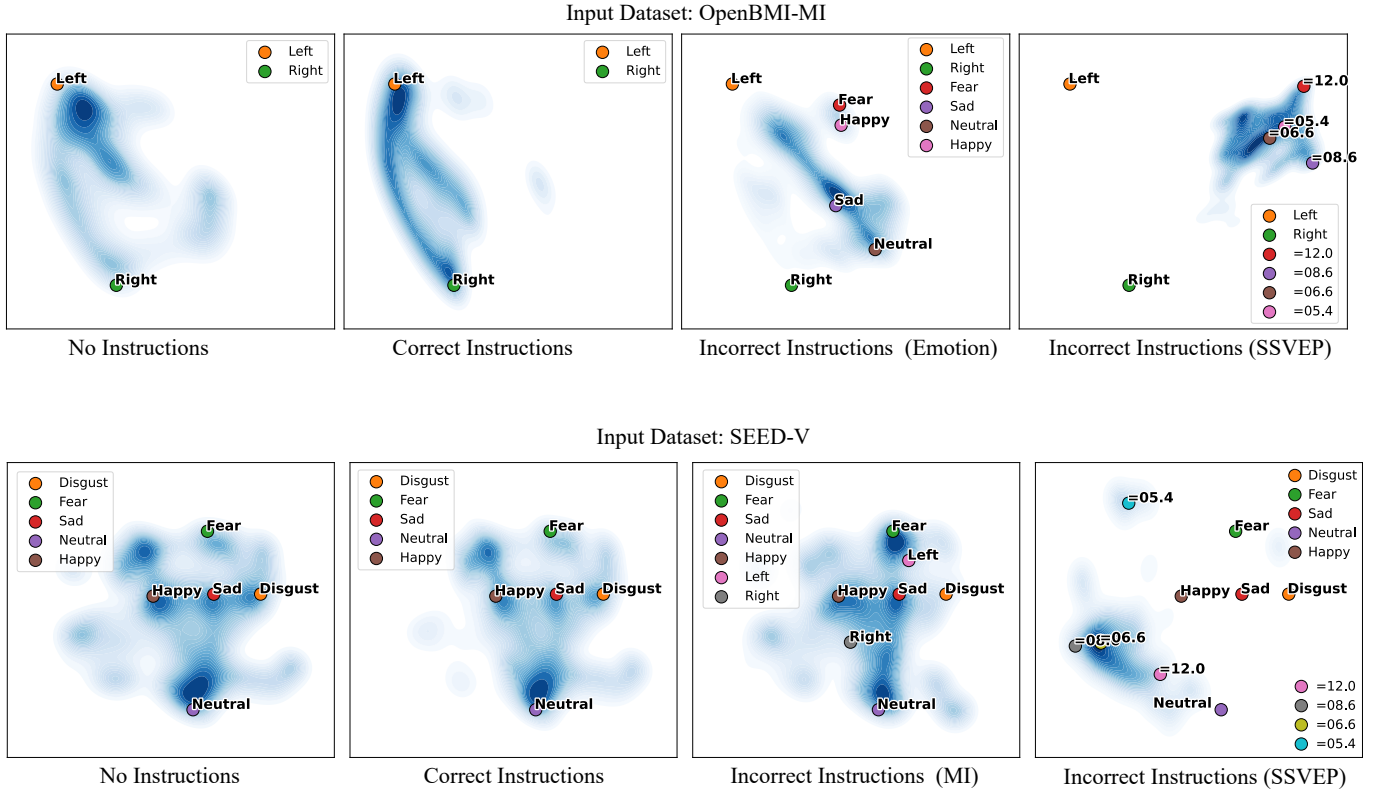


Fig. 9: Comparison of KDE visualization of features between incorrect and correct instructions.

- Chisco dataset [23]: The Chisco dataset is a large-scale EEG corpus collected from three subjects for imagined speech decoding, featuring over 20,000 sentences and more than 900 minutes of high-density EEG per subject. It covers 6,000+ everyday phrases across 39 semantic categories, with trials designed to include both reading and imagined speech phases.
- LargeSpanish [24]: This dataset consists of 60 EEG sessions from 56 healthy participants. It is recorded using a 64-channel EEG system during speech perception and silent speech production tasks involving 30 daily-use sentences in Spanish.
- ThinkOutLoud [25]: This open-access EEG dataset comprises recordings from 10 participants, collected using a 136-channel system across three paradigms—inner speech, pronounced speech, and visualized condition.

APPENDIX D

MORE DETAILS ABOUT EXPERIMENTAL SETTING ON DOWNSTREAM DATASETS

- BCIC-IV-2a dataset [27] comprises recordings from nine subjects, each participating in two sessions of a four-class MI paradigm (left hand, right hand, foot, and tongue). EEG was collected using 22 scalp electrodes and three EOG channels at 250 Hz.
- OpenBMI-MI dataset [26] provides a large-scale benchmark for brain-computer interface research. Its MI subset contains data from 54 subjects, each participating in two sessions. Subjects performed left- and right-hand motor

imagery tasks, with approximately 100 trials per session, recorded using a 64-channel EEG system at 1000 Hz.

- BCIC-Upperlimb dataset [28] is from BCI Competition 2021 – Track 4. It provides EEG recordings of subjects performing three unilateral grasp movements (cylindrical, spherical, lumbrical) across three consecutive days (train/validation/test), designed to evaluate upper-limb movement decoding and session-to-session transfer.
- SHU-MI dataset [45] includes high-quality multi-day recordings from 62 participants. Fifty-one subjects performed a two-class MI paradigm (left vs. right hand grasping), while eleven subjects performed a three-class paradigm (left hand, right hand, and foot). Each participant contributed three sessions, with both raw and preprocessed EEG data publicly available.
- High-Gamma dataset [30] was collected at TU Berlin and contains 128-channel EEG recordings from 14 subjects. Participants performed four tasks (left hand, right hand, both feet, and rest). Each subject completed 13 runs, yielding approximately 1000 four-second trials. We select left and right MI as evaluation tasks.
- Cho2017 dataset [31] contains EEG recordings from 52 subjects performing four-class motor imagery tasks (left hand, right hand, foot, tongue) using a 62-channel montage at 1,000 Hz sampling rate.
- PhysioNet-MI [33] is a publicly available dataset on PhysioNet. It comprises EEG recordings from 109 healthy subjects performing both motor execution and motor imagery tasks involving the left and right hands.

- Shin2017A [32] dataset contains EEG recordings from 30 healthy subjects (29 right-handed, 1 left-handed; average age 28.5 ± 3.7 years). Subjects performed two-class hand motor imagery tasks (left vs. right hand) using a 30-channel EEG montage at 1000 Hz. Each participant completed three sessions with 20 trials per session (10 per class).
- SEED dataset [34] contains EEG and eye movement data of 12 subjects and EEG data of another 3 subjects. Data was collected when they were watching film clips.
- SEED-IV [35] contains data from 15 subjects, each undergoing three sessions. During each session, 24 movie clips were used to elicit four discrete emotions: happy, sad, fear, and neutral. EEG was recorded using a 62-channel NeuroScan system at 1000 Hz, along with synchronized eye-tracking signals.
- SEED-V [36] expands the categories to five (happy, sad, fear, disgust, and neutral) and includes recordings from 20 subjects, each with three sessions and 15 clips per session.
- SEED-VII [37] further extends the label space to seven categories (happy, sad, fear, disgust, neutral, anger, and surprise). The dataset uses 80 video stimuli and includes EEG together with Tobii Pro Fusion eye-tracking from 20 subjects.
- FACED [38] includes EEG recordings from 123 healthy participants exposed to 28 film clips designed to induce nine fine-grained emotions: amusement, inspiration, joy, tenderness, anger, fear, disgust, sadness, and neutral. EEG was collected using a 32-channel cap (10–20 system) at 250 Hz. In addition to categorical labels, dimensional ratings such as valence, arousal, familiarity, and liking were provided.
- OpenBMI–SSVEP [26] is part of the OpenBMI dataset and provides recordings from 30 healthy adults across two sessions. The paradigm consisted of 4 visual targets presented at the screen edges, with stimulation frequencies of 5.45, 6.67, 8.57, and 12 Hz. The dataset is well-suited for studying low-frequency responses, small-class classification, and cross-session robustness.
- BCIC2020-3 dataset [28], released as part of the International BCI Competition 2020, contains multi-class imagined speech EEG recordings from 15 healthy subjects. Participants were instructed to imagine speaking five short phrases while 64-channel EEG signals were recorded, providing a benchmark resource for covert speech decoding research.
- Mental Workload [39] contains 36 subjects performing serial subtraction. EEG was recorded using 19 electrodes at a sampling rate of 500 Hz.

to a fixed length of 200 samples (65 channels), and, for transformer-based foundation models, further segmented into non-overlapping 200-sample windows as tokens. All models were trained for 100 epochs using the Adam optimizer with a learning rate of 1×10^{-3} and cross-entropy loss. We evaluated balanced accuracy, ROC-AUC, weighted F1, and Cohen’s Kappa on the validation and test splits. Pretrained checkpoints of LaBraM, EEGPT, and CBraMod were loaded and fine-tuned end-to-end from their official releases; for LaBraM and NeuroLM, we adopted the base-model variant. Because NeuroLM is a generative model without a straightforward task-specific fine-tuning protocol, a direct comparison in the task-specific finetuning setting is not applicable. We therefore compare LEAF with NeuroLM exclusively under the direct inference setting, where LEAF is evaluated after multi-task instruction tuning without any further fine-tuning.

APPENDIX E

DETAILS ABOUT BASELINE IMPLEMENTATION

To ensure a fair comparison, all baseline models were re-trained or fine-tuned using their officially released implementations and recommended hyperparameters. For each dataset, EEG trials were resampled to 200 Hz, truncated